

Project ID :

1. Topic (12 words max)

AI-Augmented Methods for Interpretable and Efficient Modeling in Modern Astronomy

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

In recent years, data science has seen rapid advances in areas such as deep learning, self-supervised learning, and explainable artificial intelligence (XAI). These developments are now mature enough to support interpretable and efficient modeling in scientific fields like astronomy, where both accuracy and transparency are essential. Traditionally, astronomers have been cautious about adopting black-box AI models, as the goal is not only to predict phenomena but to understand the physical processes behind them. However, with the emergence of physics-informed neural networks (PINNs) [1] and interpretable machine learning methods [2], there is growing confidence in using AI not just for automation but for discovery.

Astronomy is inherently data centric. Because we cannot physically interact with most celestial objects, our understanding depends on light collected by telescopes and transformed into structured datasets — images, spectra, and time-series observations. Modern surveys like SDSS, ZTF, and Pan-STARRS generate massive, high-dimensional datasets that are often incomplete, noisy, or unlabeled [3]. This makes astronomy a prime application area for AI techniques such as self-supervised learning [4], contrastive learning [5], and attention-based models for time-series data [6].

This project explores how modern AI techniques can be integrated into four key astronomical challenges. Physics-informed neural networks with XAI are used to simulate galaxy formation while making model decisions explainable. Attention-augmented RNNs uncover variability patterns in quasars. Contrastive learning improves classification of rare astronomical transients. Self-supervised learning is applied to infer stellar population properties in faint, distant galaxies. Each method reflects how AI can now support not only efficient computation but also interpretable, scientifically-aligned modeling in modern astronomy.

References

1. Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). *Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations*. Journal of Computational Physics, 378, 686–707. <https://doi.org/10.1016/j.jcp.2018.10.045>
2. Samek, W., Montavon, G., Vedaldi, A., Hansen, L. K., & Müller, K. R. (2021). *Explainable AI: Interpreting, explaining and visualizing deep learning*. Springer Nature.
3. Ivezić, Z., et al. (2019). *LSST: From Science Drivers to Reference Design and Anticipated Data Products*. The Astrophysical Journal, 873(2), 111. <https://doi.org/10.3847/1538-4357/ab042c>
4. Chen, T., Kornblith, S., Norouzi, M., & Hinton, G. (2020). *A simple framework for contrastive learning of visual representations*. ICML.
5. Abdurrahman, M., & Ishida, E. E. O. (2022). *Semi-supervised learning with contrastive pretraining for photometric classification of astronomical transients*. Monthly Notices of the Royal Astronomical Society, 514(2), 2713–2727.
6. Hložek, R., et al. (2020). *Learning to detect unusual quasars with recurrent neural networks*. MNRAS, 497(4), 5116–5134.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

This research leverages advancements in data science to design interpretable, efficient AI systems that address complex modeling and analysis challenges in astronomy. The project is structured into four modular components, each targeting a specific task in observational or simulated astrophysics, unified by the emphasis on explainability and performance.

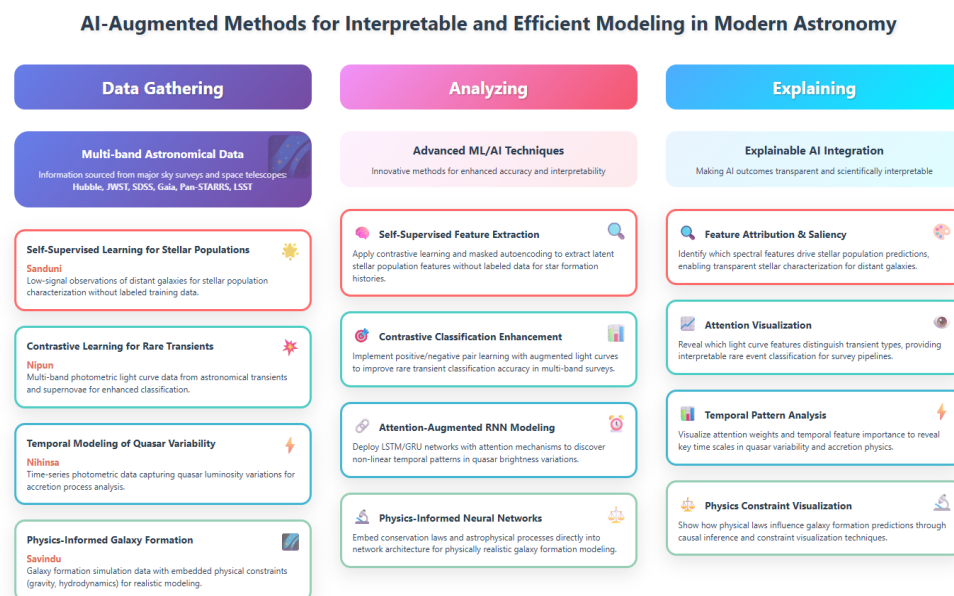
1. **Simulation of Galaxy Formation with Physics-Informed Neural Networks (PINNs):**
We develop a physics-constrained neural network that embeds known astrophysical laws into its learning process. Integrated XAI techniques are used to interpret model decisions and identify key input factors driving simulated galaxy evolution.
2. **Quasar Variability Modeling using Attention-Augmented RNNs:**
This component applies sequence models enhanced with attention mechanisms to uncover non-linear brightness variations in quasars. The model captures temporal dependencies in light curves, supporting insights into active galactic nuclei behavior.
3. **Contrastive Learning for Classification of Rare Astronomical Transients:**
Using self-supervised contrastive learning, we improve the separation of feature

representations in light curve data. This helps identify rare transient with limited labeled examples.

4. **Stellar Population Characterization via Self-Supervised Learning:**

By extracting latent features from faint and distant galaxy data, this module predicts stellar population properties without requiring labeled training sets, making it suitable for early-universe studies.

Each component includes explainable outputs to ensure transparency, with models evaluated not just for accuracy, but also for physical consistency and interpretability — enabling responsible and insightful AI use in astronomy.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This research brings together advanced data science techniques and modern astronomical data to develop interpretable and efficient AI models for astrophysical analysis and simulation. To carry out this work effectively, a combination of specialized expertise, domain knowledge, and access to suitable datasets is required.

Knowledge in data science and machine learning:

Researchers must have a strong foundation in machine learning, deep learning, self-supervised and contrastive learning, time-series modeling, and physics-informed neural networks. Familiarity with explainable AI (XAI) techniques is crucial to ensure the interpretability of models developed for scientific use. Proficiency in Python and libraries such as PyTorch, TensorFlow, Scikit-learn, and XAI frameworks is essential for implementing and evaluating the models.

Understanding of astronomy and astrophysics principles:

Basic knowledge of astrophysical processes is needed to guide model design and evaluate outputs in physically meaningful ways. This includes an understanding of galaxy formation and evolution, quasar activity, transient phenomena, and stellar population synthesis. The ability to interpret astronomical data formats (e.g., light curves, spectra, simulations) is necessary to align model inputs with relevant physical concepts.

Access to large-scale astronomical datasets:

This research relies on high-quality, multi-source datasets including simulations and observations. Key sources include the Sloan Digital Sky Survey (SDSS), Zwicky Transient Facility (ZTF), Pan-STARRS, and cosmological simulation datasets such as IllustrisTNG and EAGLE. These provide light curves, photometric catalogs, and time-resolved simulation data critical for training and validating each model.

By combining computational expertise with domain understanding and leveraging publicly available astronomical datasets, this project aims to produce trustworthy and generalizable AI systems for real-world use in astronomy.

8. Objectives and Novelty

Main Objective

- Design a physics-constrained neural network (PINN) with integrated explainable AI methods to simulate galaxy formation and uncover how physical inputs influence modeled outcomes.
- Model quasar luminosity variability using attention-augmented recurrent neural networks (RNNs) to capture non-linear time-series patterns relevant to black hole behavior.
- Apply contrastive learning to multi-band astronomical light curves to improve classification of rare and diverse transient events using self-supervised feature separation.
- Use self-supervised learning to derive informative representations of faint and distant galaxies, enabling prediction of stellar population properties without labeled data.

- Address interpretability, data sparsity, and generalization in astronomical modeling by combining physics knowledge with modern deep learning architectures. - Benchmark the performance, explainability, and robustness of these AI systems against traditional simulation and classification approaches in astronomy.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Sihilel D.G.S. IT22569004 <i>Component 1</i>	1. Build a neural network that enforces physical laws through loss constraints	1.1 Identify key physical equations relevant to galaxy formation (e.g., Navier-Stokes, gravitational potential). 1.2 Reformulate these equations as loss components usable in neural network training. 1.3 Design and train a base PINN architecture using selected constraints.	Translate galaxy formation physics into neural loss terms, enabling the model to simulate complex processes without direct supervision, reducing overfitting and increasing physical consistency.
	2. Train the PINN using galaxy formation simulation data across varying scenarios	2.1 Collect and preprocess galaxy simulation data from sources like IllustrisTNG and EAGLE. 2.2 Define model inputs/outputs: e.g., initial density fields to final halo mass distributions.	Demonstrate that a physics-guided network can generalize across simulation regimes — reducing data dependency and enabling use with incomplete or noisy inputs.

		2.3 Train the PINN on this data, testing its generalization across galaxy sizes, types, and redshifts.	
	3. Implement model explainability to identify key drivers of galaxy formation	3.1 Apply explainability tools (e.g., saliency maps, SHAP, gradients) to inspect learned representations. 3.2 Visualize which input features or physical terms most influence simulation outcomes. 3.3 Correlate these insights with known astrophysical phenomena (e.g., influence of density perturbations).	Enable insights into how the model prioritizes physical inputs, offering traceable and interpretable mappings that connect input conditions to simulated galaxy formation outcomes.
	4. Benchmark the interpretability and accuracy of the PINN against black-box models	4.1 Define evaluation metrics for both performance (error, generalization) and interpretability (clarity, fidelity). 4.2 Compare the PINN with conventional deep networks and traditional simulators. 4.3 Analyze trade-offs and produce a comparative summary.	Present a data-driven benchmark showing that interpretable, physics-guided models can rival or surpass standard models—supporting transparent, scientifically aligned AI use in cosmology.
IT22133922 – Senevirathna B.A.N.K <i>Component 2</i>	1. Develop a contrastive learning framework to extract feature representations	1.1 Conduct a literature review on contrastive/self-supervised learning in astronomy.	Introduce a contrastive learning pipeline for sky survey data, reducing dependency on labeled samples. Tailors augmentations and encoder

	from multi-band light curves of transient astronomical events.	1.2 Collect and preprocess multi-band transient light curves using relevant data sets. 1.3 Design suitable augmentations for contrastive learning. 1.4 Train a suitable model to learn generalizable feature embeddings.	design specifically for transient light curves.
	2. Fine-tune the pretrained encoder using limited labeled samples to classify rare astronomical transients.	2.1 Acquire few-shot labeled data for key transient types (supernovae, kilonovae, etc.). 2.2 Add classification head and fine-tune model on labeled samples. 2.3 Evaluate model performance using precision, recall, and F1-score. 2.4 Analyze performance in class-imbalanced settings.	Apply a classification strategy using contrastively learned embeddings to improve detection of rare classes, outperforming traditional supervised models.
	3. Evaluate the contrastive learning approach against supervised baselines in	3.1 Train baseline models directly on labeled data. 3.2 Visualize learned embeddings using different method.	Provide a novel comparative analysis of contrastive learning vs. traditional supervised learning in astronomy. Emphasizes benefits for representation quality, generalization, and training with limited data.

	terms of representation quality and classification performance.	3.3 Compare metrics: accuracy, confusion matrix, generalization on unseen transients. 3.4 Document advantages in training efficiency and class robustness.	
Dilmani H.V.N IT22122032 <i>Component 3</i>	1. Build an RNN to learn how quasar brightness changes over time	1.1 Collect and preprocess quasar light curves from surveys. 1.2 Build and train an RNN model to learn how brightness changes over time. 1.3 Use metrics to check model accuracy and prevent overfitting. 1.4 Analyze how brightness model properties of quasars	Moving beyond traditional spectral line fitting: Traditional spectroscopy often relies on manual or semi-automated fitting of emission and absorption lines to estimate quasar properties like redshift, black hole mass, and accretion rate. These methods are precise but can be time-consuming, sensitive to noise, and limited in handling complex or blended features.
	2. Include basic physical rules in the model's training	2.1 Identify key physical rules (e.g., energy conservation, known variability timescales). 2.2 Add these rules into the model's loss function as constraints.	Combining spectral ML models with physical constraints. Adding physics rules to ML models helps understand their results better and connect them to real quasar behavior

		2.3 Make sure the model follows these rules while learning from the data. 2.4 Test if adding physical rules improves accuracy and interpretability.	
	3. Increase model accuracy by validating predictions with traditional astrophysical methods	3.1 Select traditional variability analysis methods (e.g., structure function, autocorrelation) relevant to quasar light curves. 3.2 Apply these methods to the same quasar light curves used in the model. 3.3 Compare the model's predicted variability patterns with results from traditional methods to assess alignment and accuracy. 3.4 Interpret results to connect model behavior with physical properties of quasars	This approach improves model accuracy by comparing deep learning predictions with results from traditional variability analysis methods, while also identifying which parts of the light curve influence predictions.
Gunasinghe - W M S H IT22205056 <i>Component 4</i>	1. Learn representative features from	1.1 Collect and preprocess galaxy data (e.g., spectra and images) from public databases or simulations. 1.2 Design a method that learns useful	Use a self-supervised learning approach to extract meaningful features from galaxy data without relying on labeled samples, which is rare in astrophysical studies. Embeds

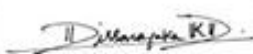
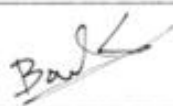
	unlabeled galaxy data	patterns from the data without using labels. 1.3 Introduce domain knowledge to guide the learning process. 1.4 Train the model to generate reliable features, even when the input data is noisy or incomplete.	physical knowledge and domain-specific constraints directly into the learning process, ensuring that learned features are not only data-driven but also scientifically valid.
	2. Implement disentangled self-supervised learning for feature separation	2.1 Apply data transformations that reflect different stellar populations (e.g., metallicity, mass). 2.2 Train the model to learn and separate these characteristics into meaningful and independent features. 2.3 Evaluate how well the features represent each physical property using visual and statistical methods.	Creates a method to separate different physical properties of stellar populations into distinct and understandable features. This helps in better analyzing and interpreting complex galaxy characteristics from observational data.

	3. Predict stellar population properties with limited labeled data	3.1 Collect a small set of labeled galaxy data for selected physical properties. 3.2 Use the previously learned features to make predictions on these properties. 3.3 Measure prediction accuracy using appropriate evaluation techniques. 3.4 Assess how well the predictions hold up when working with small or noisy datasets.	Demonstrates that features learned through self-supervised learning can be effectively reused for prediction tasks with very minimal labeled data, addressing a major limitation in astronomy research.
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9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Sihilel D.G.S. IT22569004	This component exemplifies the application of data science in a scientific domain. It integrates deep learning, explainable AI, and physical modeling to develop data-efficient, interpretable models of galaxy formation. By embedding physical laws into the learning process, the project demonstrates how data science can extend beyond prediction—toward building models that are both accurate and scientifically valid.
Senevirathna B.A.N.K IT22133922	This research aligns with the Data Science specialization by applying contrastive learning to classify rare astronomical transients using sky survey data. It uses advanced frameworks to learn from unlabeled light curves, addressing the scarcity of labeled data in astronomy. Through tailored data augmentation and fine-tuning, the model achieves efficient classification of rare events. The study demonstrates core data science techniques—representation learning, deep learning, and statistical evaluation—and contributes to the field of astro informatics, bridging AI methods with real-world astronomical data challenges.
Dilmani H.V.N IT22122032	This component uses deep learning to study how the brightness of quasars changes over time. This will build a neural network with an attention mechanism that focuses on important moments in their light curve data. This will help the model make accurate predictions and highlight which time periods are most relevant to quasar variability. I will work with real astronomical time-series data, apply data preprocessing techniques, train the model, and analyze its results to uncover patterns in quasar behavior.
Gunasinghe W.M.S.H IT22205056	This component applies data science to astronomy by developing methods that help computers learn useful information about galaxies without needing a lot of labeled data. It uses raw galaxy data and scientific knowledge to teach the system to recognize important patterns. These patterns are then separated into meaningful features, such as metallicity or mass, making them easier to understand. Finally, the learned features are used to predict galaxy properties, even with limited labeled examples. This approach helps study and classify stellar populations in distance galaxies more effectively, while ensuring the results are scientifically accurate.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Samadhi	Rathnayake	
Co-Supervisor	Dr.	Kapila	Dissanayake	
External Supervisor	Mr	Banuka	Kularatne	
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	
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- b) Does the proposed topic exhibit novelty?

Yes		No	
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

[illegible]

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.